1. Network Discovery Using Nmap

- Nmap (Network Mapper) is a network-scanning tool used to discover devices connected to a network.
- Install Nmap from the official download page: **https://nmap.org/download.html**. Open Command Prompt or Terminal after installation.
- Identify the local network range. On Windows, run **ipconfig** and check the IPv4 address and subnet mask. On Linux, run **ip route**.
- Run a host-discovery scan on your local network. Replace the example subnet with your actual network range.
- The **-sn** option discovers active hosts without scanning ports, and **-oN** saves the output in a text file.

Nmap • Host Discovery
Command
nmap -sn 192.168.1.0/24 -oN scan_results.txt
Command explanation
• -sn : discover active hosts without scanning ports
• 192.168.1.0/24 : example local network range
• -oN scan_results.txt : save the output to a text file
Result
Nmap displays the active hosts detected on the selected network. The output is saved in scan_results.txt.

2. Reconnaissance of a Popular Application

- Application selected: **Instagram**.
- Reconnaissance is the information-gathering phase of penetration testing. It helps a tester understand the application before security testing.
- A tester may review Instagram's official website, public application information, login and account-recovery workflows, and documented features.
- Other information may include public domains and endpoints, supported platforms, visible technologies, security-related response headers, and public security advisories.
- Major application areas that may be reviewed include authentication, profile management, messaging, and media uploads.
- This information helps identify the application's public attack surface and prepare an authorized testing plan.

Instagram • Reconnaissance Summary
Application: Instagram
Information gathered
• Official website and public application endpoints
• Login and account-recovery process
• Publicly documented features and supported platforms
• Visible technologies and security-related response headers
• Public security advisories and vulnerability information
Outcome
The information helps the tester understand the application's public attack surface and plan further security testing.

3. Port Scanning of a Public Test Server

- A port scan checks the state of network ports on a target host. Ports may be reported as open, closed, or filtered.
- For this practical, use **scanme.nmap.org**, a test server provided by the Nmap project for learning and testing Nmap.
- Open a terminal with the required permissions and run the TCP SYN scan command shown below.
- The **-sS** option performs a TCP SYN scan. Nmap analyzes the responses to identify the state of TCP ports.
- The **-oN** option saves the scan results in a text file. Review the PORT, STATE, and SERVICE columns to identify open ports.

Nmap • TCP SYN Port Scan
Command
nmap -sS scanme.nmap.org -oN public_scan_results.txt
Command explanation
• -sS : perform a TCP SYN scan
• scanme.nmap.org : Nmap's test server
• -oN public_scan_results.txt : save the scan output
Open ports
Check the STATE column and record the port number, protocol, and service for each entry marked open.
Output file: public_scan_results.txt

4. Client-Style Scan Report

- A scan report summarizes the purpose, method, findings, and recommendations for a client or network administrator.
- The local network discovery scan uses Nmap to identify active devices on the selected local subnet.
- The TCP SYN scan uses the Nmap test server, scanme.nmap.org, to identify reachable TCP ports and their services.
- An open port indicates that a service is reachable. It does not automatically mean that the service is vulnerable.
- Potential risks may include unnecessary exposed services, outdated software, or weak service configurations.
- Recommended actions include verifying exposed services, applying security updates, restricting unnecessary access, and reviewing service configurations.

Basic Network Scan • Client Report
To: Client / Network Administrator
Subject: Basic Network Discovery and Port Scan Report
Scope: Local network and scanme.nmap.org
Method: Nmap host discovery (-sn) and TCP SYN port scanning (-sS)
Findings: The host-discovery scan identifies active devices on the selected local subnet. The TCP SYN scan identifies the state of ports detected on the test server.
Potential risks: Unnecessary exposed services, outdated software, or insecure configurations may increase the network's attack surface.
Recommendations: Verify exposed services, restrict unnecessary access, apply updates, and review service configurations.
Conclusion: Nmap helps identify active hosts and reachable network services. The scan findings can be used to guide further security review.